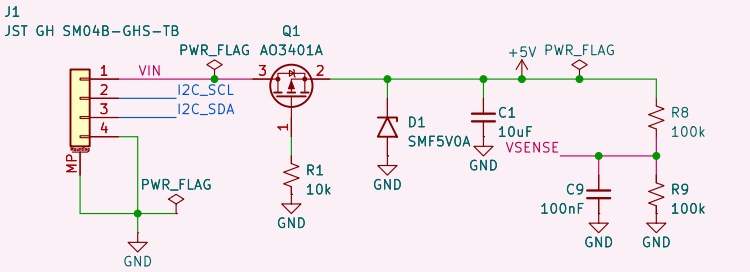
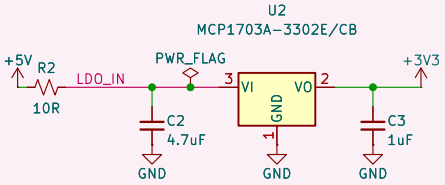
POWER INPUT, PROTECTION, SUPPLY SENSE



VSENSE = +5 V / 2 (PA4)

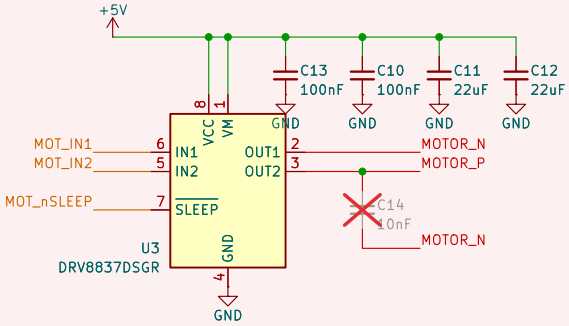
J1: JST GH 1.25 mm, Dronecode/Pixhawk I2C pinout: 1 = +5 V, 2 = SCL, 3 = SDA, 4 = GND

3.3 V LDO



10R + 4.7uF: keeps motor PWM ripple out of the LDO.
MCP1703A: 16 V max in, 2 uA Iq

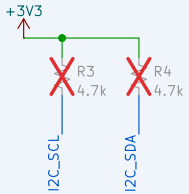
MOTOR DRIVER (DRV8837, motor runs from +5 V)



C13 = VCC, C10–C12 = VM decoupling. C14 not fitted (brush-noise option).
IN2 = PWM, IN1 = 0: M+ positive, knob moves to the top of travel (ALPS).
IN1 = PWM, IN2 = 0: knob moves down. IN1 = IN2 = 1: brake.

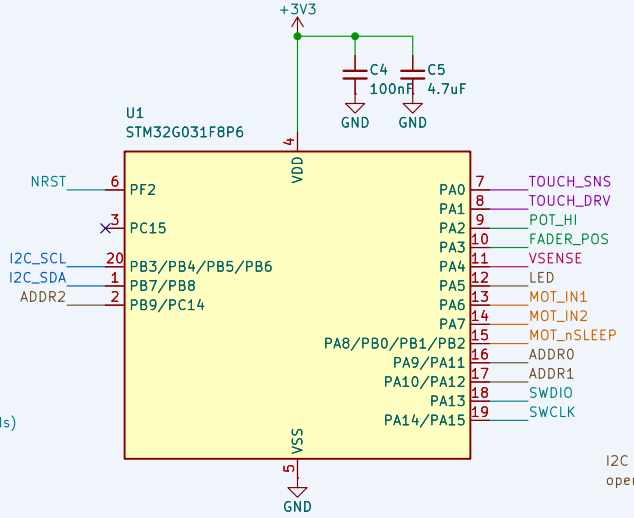
MCU STM32G031F8P6

I2C pull-ups: fit on one board per bus only

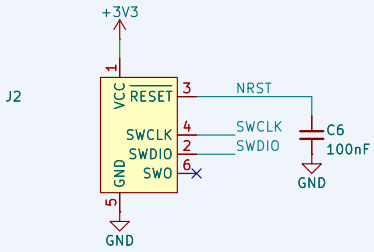


Status LED

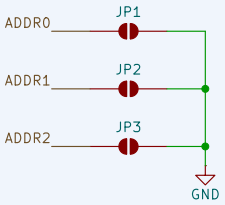




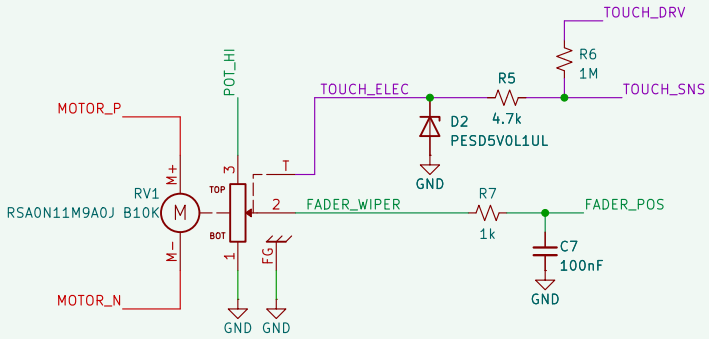
SWD (Tag-Connect TC2030-NL pads)



I2C address select:
open = 1, bridged = 0



FADER: POSITION, TOUCH, MOTOR (ALPS RSA0N11M9A0J)



Touch: conductive knob -> lever -> T. Sense by RC charge time (PA1 drives via 1M, TIM2 capture on PA0) or by ADC charge sharing on PA0.

Position: track powered from POT_HI (PA2 high), wiper read on PA3 (ratiometric). End 1 (bottom of travel) = GND, so FADER_POS rises toward the top.

FG = fader frame lugs: soldered for retention, tied to GND.

STM32G031F8P6, DRV8837 H-bridge, MCP1703A 3.3 V LDO.
Fader: ALPS RSA0N11M9A0J (clone B0H45B29YK). 5 V + I2C on JST GH 4-pin.

Sheet: /
File: E0001_fader_driver.kicad_sch

Title: E0001 Motorized fader driver board

Size: A4	Date: 2026-10-06	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1